Potential Wages and the Value of a Domestic Credential: A Simulated Maximum Likelihood Approach for Foreign Workers in South Korea

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Abstract

This paper investigates the economic returns to a host-country (domestic) degree for immigrants in South Korea, a non-traditional and ethnically homogeneous immigration context where the value of such credentials may not align with findings from Western economies. Using pooled cross-sectional data from the Survey on Immigrants' Living Conditions and Labor Force (2017-2022), we employ a sample-selection corrected Ordered Probit model with instrumental variables to simultaneously address the endogeneity of educational choice, sample selection into employment, and the ordinal nature of wage data. Our findings reveal a nuanced, dynamic relationship. The direct wage effect of a domestic degree is statistically insignificant for immigrants with little to no host-country work experience. However, we find a strong, positive, and significant interaction between holding a domestic degree and accumulating three or more years of work experience. The analysis of Average Partial Effects (APEs) confirms that for experienced workers, a domestic degree significantly decreases the probability of remaining in lower-wage categories while increasing the probability of securing high-wage employment. These results suggest that a domestic degree in Korea acts as a latent asset whose economic value is unlocked or activated over time through on-the-job experience, highlighting the critical interplay between formal credentials and practical workplace integration.

- *Keywords*: Foreign-born Workers, Host-Country Education, Sample Selection
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1 Introduction

The accelerating pace of globalization has rendered international migration a defining feature of the twenty-first century, fundamentally reshaping the demographic and economic landscapes of nations across the globe. As capital and goods flow across borders with increasing ease, so too do people, seeking economic opportunity, refuge, or new life experiences. This sustained movement of labor, particularly from lower to higher-income countries, is not a series of unprecedented crises but rather a steady, persistent force that has been operating for over half a century. The economic integration of these immigrant populations into their host societies is not merely a matter of individual success or failure; it is a critical determinant of a nation's overall economic efficiency, social cohesion, and long-term prosperity. A central question that has animated decades of academic and policy debate is how immigrants fare in their new homes and, more specifically, what factors facilitate their successful incorporation into the host country's labor market. A vast body of literature, originating with the seminal work of scholars like Chiswick (1978) and Borjas (1985), has established a foundational narrative of immigrant assimilation. This narrative posits that while immigrants typically face an initial earnings disadvantage compared to native-born workers, their wages tend to grow at a faster rate over time, leading to a gradual convergence. The primary engine of this economic "catch-up" is understood to be the accumulation of human capital, particularly skills and knowledge that are specific to the host country. A core tenet of this research is the concept of the limited "transferability of human capital". Education, professional skills, and work experience acquired in an immigrant's home country are often heavily discounted in the new labor market, a phenomenon attributed to variations in educational quality, differences in institutional and cultural norms, and language barriers. Consequently, human capital acquired within the host country—be it local work experience or a formal educational credential—is expected to command a significant wage premium, serving as the most reliable vehicle for economic advancement. This established framework naturally leads to the hypothesis that a domestic degree should be a powerful asset for any immigrant. It represents not only the acquisition of tangible, locally relevant skills but also a credible signal of an individual's ability, perseverance, and commitment to adapting to their new society. However, the vast majority of this foundational research has been conducted within the context of traditional Western settlement nations, such as the United States, Canada, and Australia. These countries share long histories of large-scale immigration and have developed, to varying degrees, multicultural frameworks and institutions to manage diverse populations. The question remains as to whether these findings can be universally applied to nations with vastly different historical, cultural, and institutional trajectories. South Korea presents a unique and compelling case for re-examining these established theories. Over the past few decades, Korea has undergone a dramatic transformation, shifting from a nation of emigration to a significant destination for immigrants, with its foreign-born population increasing at one of the fastest rates among OECD countries. This rapid influx has occurred within a society long characterized by a high degree of ethnic and linguistic homogeneity. As Korea navigates its transition into a more multicultural society, an empirical understanding of immigrant integration is not just an academic exercise but a pressing policy imperative. Yet, empirical research on this topic remains surprisingly limited. While the general assumption might be that a Korean university degree would confer a substantial advantage to a foreign resident, this has not been rigorously tested. The signaling value of a credential is not inherent; it is interpreted by employers within a specific social and cultural milieu. It is entirely plausible that in the Korean context, the institutional mechanisms and social perceptions that shape labor market outcomes operate differently than in the West. Studies in other contexts, such as Støren and Wiers-Jenssen's (2010) finding in Norway that a non-Western immigrant's ethnic background was a more powerful determinant of labor market mismatch than the origin of their diploma, provide a precedent for questioning the universal efficacy of domestic degrees. Therefore, an urgent need exists to move beyond theoretical assumptions and empirically investigate the actual economic value of a domestic degree within the specific context of the South Korean labor market. This study aims to fill this critical gap, providing a nuanced understanding of immigrant integration in a new and important global context. Theoretically, the case for a domestic degree premium rests on two robust pillars: human capital theory and signaling theory. From a human capital perspective, the process of obtaining a domestic degree is an intensive investment in host-country-specific skills that directly enhance a worker's productivity. It goes far beyond the subject matter of the degree itself. It necessitates the mastery of the local language to an academic and professional level, a skill with proven, substantial returns on earnings. It forces an immersion in the host country's institutional and bureaucratic systems, from university administration to professional etiquette. It helps build invaluable social and professional networks that are often crucial for finding employment and advancing in a career. In essence, a domestic education transforms an immigrant's skill set, making it more legible and valuable to local employers, which, according to human capital theory, should be rewarded with higher wages. From a signaling theory perspective, a domestic degree is perhaps even more critical. Employers, especially when hiring immigrants, operate under conditions of significant information asymmetry. They may have little ability to verify or accurately assess the quality of foreign credentials, the relevance of foreign work experience, or the unobserved characteristics of a foreign applicant. A degree from a recognized domestic institution acts as a powerful and credible signal that cuts through this uncertainty. It signals not only cognitive ability and domain-specific knowledge but also a host of other desirable traits: perseverance, the ability to navigate a challenging new environment, and a demonstrated level of social and cultural assimilation. As Ferrer and Riddell (2008) argued in their work on "sheepskin effects," the premium for a completed credential can be even higher for immigrants than for natives precisely because the signal is so much more valuable in an otherwise information-poor environment. The diploma is a standardized, trusted currency in a market where foreign qualifications are often viewed as foreign currency of uncertain value. Despite this powerful theoretical foundation, the actual realization of this premium in the labor market is far from guaranteed. The journey from a university graduation stage to a higher-paying job is fraught with real-world frictions that can devalue or even nullify the credential's theoretical worth. The most significant counterforce may be the primacy of social identity over individual qualifications. Labor markets are not frictionless, perfectly rational arenas; they are social structures embedded with biases, stereotypes, and in-group preferences. An employer's decision-making can be heavily influenced by the coarse, macro-level identity of "foreigner," which can easily overshadow the finer, micro-level signal of a "domestic degree holder." This can manifest as taste-based discrimination, where an employer simply prefers not to hire foreigners, or as statistical discrimination, a more subtle process where employers use perceived group averages to judge an individual. If an employer harbors a preconceived notion that foreign workers, as a group, are less productive, have higher turnover rates, or are more difficult to manage, they may assign this stereotype to every foreign applicant, regardless of their individual merits or qualifications. In such a scenario, the positive signal sent by a domestic degree is effectively muted, drowned out by the noise of prejudice. Furthermore, cultural distance and institutional barriers can act as persistent drags on the economic returns to a domestic degree. Even an immigrant fluent in the local language and holding a local degree may struggle with subtle differences in communication styles, workplace hierarchies, and unwritten rules of corporate culture, hindering their performance and chances for promotion. Institutional hurdles, such as complex visa regulations that tie workers to specific industries or difficulties in obtaining professional licenses even with a domestic credential, can also place an artificial ceiling on an immigrant's earning potential. These effects can be compounded by other social identities. As research by Arbeit and Warren (2013) has shown, immigrant women often face a "double disadvantage," confronting biases related to both their gender and their national origin, which can further erode the value of their qualifications. Ultimately, a theoretical tension exists: the final economic outcome for a domestic-degree-holding immigrant depends on which force proves stronger within a given labor market—the productivity- and signal-enhancing effect of their education, or the productivity- and signal-ignoring effect of social categorization and discrimination. To empirically adjudicate between these competing forces, this study focuses on a comparative analysis of potential wage levels among foreign residents in South Korea. Our research design partitions the foreign population into two distinct groups: those who hold a degree from a Korean educational institution and those whose highest degree was obtained abroad. The central objective is to rigorously test whether the acquisition of a domestic degree provides a statistically and economically significant wage premium, after controlling for a comprehensive set of individual and labor market characteristics. Crucially, our analysis aims to move beyond a simple comparison of observed average wages. The focus is on estimating the potential wage—that is, the expected wage an individual would command in the labor market based on their human capital and other attributes, irrespective of whether they are currently employed. Observed wages are only available for the non-random subgroup of individuals who have successfully found work, a sample that may not be representative of the entire population. By aiming to estimate potential wages, we can get closer to a true measure of the market's valuation of a domestic degree and therefore provide a more precise evaluation of its causal effect on earnings potential. This approach allows us to address the question not merely of correlation ("Do domestic degree holders in Korea earn more?") but of causation ("Does obtaining a domestic degree cause an increase in a foreigner's potential earnings?"). The null hypothesis of this study is that, after accounting for other factors, no significant difference exists in the potential wage distributions of the two groups. The methodological approach of this study is carefully tailored to the specific characteristics of the available data on foreign workers in South Korea and the inherent econometric challenges of this type of analysis. A common feature of social survey data, particularly concerning sensitive information like income, is that it is often collected in categorical form rather than as a precise, continuous variable. Our dataset is no exception; the dependent variable for wages is structured as an ordered set of quantiles or brackets. In this context, a standard Ordinary Least Squares (OLS) regression model is inappropriate, as it would impose arbitrary numerical distances between the wage categories and violate its core assumptions, leading to biased and inefficient estimates. The most suitable econometric tool for this data structure is the Ordered Probit model. This model is specifically designed for ordinal dependent variables, estimating the probability that an individual's latent wage falls within a particular category as a function of their observed characteristics. It respects the ordered nature of the data without making unwarranted assumptions about the magnitude of the differences between wage levels, thus providing a more structurally sound estimation framework. Furthermore, a critical challenge in estimating the causal effect of education on earnings is the problem of endogeneity. The decision to obtain a domestic degree is not a random event. It is likely correlated with unobserved individual attributes such as innate ability, motivation, family background, or prior exposure to Korean culture, all of which could independently influence wage outcomes. To isolate the causal effect of the degree itself, we employ an Instrumental Variable (IV) approach in conjunction with our ordered probit model. The logic of the IV strategy is to identify variables (instruments) that are strongly correlated with the endogenous decision (obtaining a domestic degree) but are not directly correlated with the outcome (potential wages), except through their effect on the education decision. For this study, we propose two key instruments: the parents' country of origin and whether the individual's pre-tertiary education took place in Korea or abroad. These variables are plausibly correlated with the likelihood of pursuing a Korean university degree. For instance, an individual who attended high school in Korea is far more likely to enroll in a Korean university. At the same time, it is reasonable to argue that these instruments satisfy the exclusion restriction; a parent's nationality, for example, is unlikely to have a direct impact on their adult child's current wage level in the Korean labor market, other than through its influence on that child's educational pathway. This IV strategy is essential for disentangling correlation from causation and obtaining a more credible estimate of the domestic degree premium. An additional layer of complexity arises from the fact that wages are only observed for individuals who are employed. This gives rise to a potential sample selection bias, a problem famously articulated by Heckman (1979). The decision to participate in the labor force is not random and is likely correlated with the same factors that determine wages. For example, if foreigners who anticipate facing significant wage discrimination choose not to seek work, then a sample consisting only of employed foreigners will be skewed towards those who, for whatever reason, expected better outcomes. Estimates based on such a selected sample will not be representative of the potential wages of the entire foreign population. To address this, our study employs a model that simultaneously accounts for both the work-or-not-work decision and the determination of the wage level. This requires estimating a complex system of joint equations. Given the technical difficulties of maximizing the likelihood function for such a model—which combines a selection equation with an ordered probit outcome equation and instrumental variables—we adopt a Simulated Maximum Likelihood (SML) estimation approach. SML is a powerful computational technique that uses simulation to approximate the complex integral in the likelihood function, allowing us to robustly estimate the parameters of the entire system. This sophisticated methodology enables us to correct for selection bias and estimate the true "potential wage" for all individuals in our sample, both working and non-working, thereby strengthening the causal interpretation of our findings. After implementing this robust econometric framework, the empirical results of our study are stark. We fail to find clear and consistent evidence that holding a domestic degree provides a significant wage premium for foreigners in the South Korean labor market. This finding stands in contrast to the predictions of standard human capital and signaling theories and diverges from the bulk of evidence from traditional Western immigrant destinations. This "null result" is not a failure of analysis but a significant finding in itself, suggesting that the institutional and cultural specificities of the Korean context may severely limit or override the expected benefits of a domestic qualification. It raises the distinct possibility that in the Korean labor market, an individual's macro-level identity as "foreigner" operates as a primary and powerful characteristic that mutes the positive signal of a domestic degree. The lack of a premium could be the result of various unseen barriers, from statistical discrimination by employers to the social closure of high-paying networks to non-natives. This result provides critical empirical evidence that when discussing the economic integration of immigrants in South Korea, factors beyond educational credentials—such as social identity, discrimination, and institutional context—may play a far more decisive role. It compels a re-evaluation of integration policies, suggesting that merely encouraging foreigners to pursue domestic education may prove insufficient if deeper, structural barriers within the labor market remain unaddressed.

2 Literature Review

The economic integration of immigrants, particularly their successful incorporation into host country labor markets, stands as a central policy challenge and a focal point of academic inquiry in contemporary societies. Decades of research have repeatedly identified a common pattern: immigrants often face an initial wage disadvantage compared to their native-born counterparts, but this gap tends to narrow over time as they reside longer in the host country, a phenomenon known as "wage assimilation". The primary engine of this assimilation process is explained through the accumulation and valuation of human capital. A particularly consistent finding within this literature is that human capital acquired in the host country—be it education or work experience—commands a significantly higher value in the labor market than qualifications obtained in an immigrant's home country. Against this backdrop, the acquisition of a "domestic degree" by an immigrant has long been considered one of the most reliable pathways to labor market success. The prevailing logic is that a domestic credential serves a dual function. It is both the medium through which an immigrant embodies country-specific human capital—such as language proficiency, an understanding of local institutions, and social networks—and a powerful signal to employers, attesting to an individual's productivity, adaptability, and cultural fit. However, contrary to this general expectation, the positive effect of a domestic degree is not always clear-cut. The purpose of this literature review is to argue, by synthesizing a wide range of theoretical and empirical studies, that the impact of a domestic qualification on immigrant labor market outcomes can be far more complex and ambiguous than is often assumed. We will begin by examining the traditional arguments supporting the value of domestic degrees. Subsequently, we will critically analyze the various factors that dilute, distort, or even nullify this effect, including the profound influence of selection biases in migration, the nuanced signaling value of credentials in the face of imperfect information, and the compounding impact of social identities such as race and gender. Through this comprehensive analysis, we will advance the hypothesis that in certain labor market contexts, particularly in non-traditional immigration countries like South Korea, the macro-level identity of "foreigner" may override the micro-level qualification of "domestic degree holder," potentially rendering the credential's positive effect negligible. This framework is intended to provide a robust theoretical foundation for the provisional research finding that, in the Korean case, the effect of a domestic degree is not readily apparent.

The Host-Country Human Capital Premium: The Conventional Case for Domestic Degrees A foundational concept in the economics of immigration is the limited "transferability of human capital". The education and experience an immigrant accumulates in their country of origin are often not fully valued in a new labor market. This devaluation stems from several sources. First, there can be real or perceived differences in the quality of educational systems. Employers may be unfamiliar with foreign institutions and thus discount the qualifications they award. Second, much human capital is inherently country-specific. Knowledge of a particular nation's legal codes, accounting standards, bureaucratic norms, and cultural customs is not directly applicable elsewhere. Finally, barriers related to language and social networks further diminish the value of home-country human capital. Because of this imperfect transferability, the human capital that immigrants acquire locally, especially a domestic degree, is expected to carry a significant premium. A large body of empirical research supports this view. A study by Bratsberg and Ragan (2002) in the United States demonstrated that immigrants with U.S. schooling not only achieve higher levels of education but also earn higher rates of return per year of schooling compared to immigrants educated abroad. This is interpreted as evidence that U.S. education either "upgrades" an immigrant's existing knowledge or "certifies" its value in a way the U.S. labor market recognizes. Research in Canada and Israel similarly finds that host-country experience yields much higher returns than foreign experience, explaining a large portion of the wage gap between immigrants and natives. In Spain, the findings are analogous: returns to education acquired in the home country are substantially lower than for Spanish education, except for immigrants from other highly developed nations. This domestic degree premium can be explained through two primary theoretical lenses. First, from a Human Capital Theory perspective, domestic education directly enhances an immigrant's productivity by equipping them with specific skills and knowledge demanded by the local labor market. This includes fluency in the dominant language, mastery of local technical standards and professional norms, and the development of collaborative skills attuned to the local work culture. In a competitive market, this increased productivity is rewarded with higher wages. Second, from a Credentialing or Signaling Theory perspective, a domestic degree serves as a crucial "signal" that conveys valuable information about an immigrant's potential productivity to employers operating with incomplete information. Employers often lack knowledge about an immigrant's background, especially the quality of foreign educational institutions they attended. In this environment of information asymmetry, a degree from a recognized domestic university acts as a credible credential, certifying that the immigrant possesses a certain level of intelligence, learning ability, perseverance, and adaptability to the host society's culture. To reduce uncertainty, employers place a premium on this clear signal, which translates into higher earnings.

The Nuances of Credentials: "Sheepskin Effects" and Their Implications To understand the value of a domestic degree more deeply, it is essential to distinguish between the quantity of education (years of schooling) and the qualitative completion of a program (degree receipt). The term "sheepskin effect" refers to the earnings premium associated with obtaining a diploma or degree, even after controlling for the number of years spent in school. It captures the value of program completion itself, separate from the value of an additional year of study. This distinction is particularly illuminating in the context of immigration. Ferrer and Riddell's (2008) research in Canada provided a seminal insight: while the years of schooling and pre-migration work experience of immigrants were valued much less than those of comparable natives, the estimated sheepskin effects for immigrants' credentials were as large as, or even greater than, those for native-born Canadians. This suggests that while employers may discount the general knowledge signaled by years of foreign education, they place significant value on the definitive signal of program completion. There are several interpretations of this finding. From a signaling perspective, a degree is a less ambiguous and more powerful signal than years of schooling, especially for an immigrant. For a native worker, an employer has access to a rich set of other signals (e.g., the reputation of their high school, local networks, cultural cues). For an immigrant, whose background may be an "information vacuum" for the employer, the formal credential of a domestic degree becomes a disproportionately important signal of quality, motivation, and successful acculturation. This finding strongly implies that employers are actively seeking reliable signals from immigrant populations and are willing to reward them, reinforcing the argument that domestic degrees should have a clear, positive effect on earnings. It suggests a mechanism through which immigrants can overcome information asymmetry and demonstrate their value in a new labor market.

Confounding Factors: Why the Domestic Degree Advantage May Disappear While the theoretical case for a domestic degree premium is strong, the observed reality is often muddled by a host of confounding factors. The simple correlation between holding a domestic degree and earning a higher wage does not necessarily imply a straightforward causal relationship. Selection biases, the overriding influence of social identity, and contextual specificities can significantly weaken or even negate the expected benefits of domestic qualifications. The Distorting Lens of Selection Bias The population of immigrants who obtain degrees in a host country is not a random sample of the overall immigrant population. The processes of who decides to study abroad, and who subsequently decides to remain in the host country, introduce significant selection biases that complicate any analysis of the returns to education. First, there is the issue of initial self-selection. Individuals who migrate for the purpose of education may be systematically different from those who migrate after completing their education at home. Student migrants may possess higher levels of ambition, greater family financial support, better cognitive skills, or a stronger orientation towards assimilation—qualities that would lead to higher earnings regardless of where they obtained their degree. A naïve comparison between immigrants with domestic degrees and those with foreign degrees risks misattributing the higher earnings of the former group to the degree itself, when it may actually reflect these pre-existing, unobserved differences. Research by Chiquiar and Hanson (2005) on Mexican migration further complicates simple models by showing that migrant selection is not always "negative" (the least-skilled migrating) but can be "intermediate," with those from the middle of the skill distribution being most likely to migrate. This demonstrates the difficulty in predicting the direction of selection bias without detailed data from both sending and receiving countries. Second, and perhaps more profoundly, is the bias introduced by selective out-migration (or return migration). Cross-sectional data, such as a single census snapshot, can be deeply misleading because it only captures the immigrants who have remained in the host country at a specific point in time. If less-successful immigrants are more likely to return to their home country, then tracking a cohort across different census years will create an illusion of rapid wage growth, as the average earnings of the remaining, more successful "survivors" will rise. Lubotsky's (2007) landmark study, using longitudinal Social Security data to follow the same individuals over time, provided powerful evidence of this phenomenon. He found that the true earnings assimilation of immigrants who remain in the U.S. is roughly half as fast as that estimated from repeated cross-sections. This "survivorship bias" has direct implications for assessing the value of a domestic degree. If immigrants who earn a domestic degree are also more likely to be successful and thus more likely to remain in the host country, the observed high earnings for this group will be an overestimate of the true causal effect of the degree. The data will be skewed by the absence of the less-successful domestic-degree holders who chose to leave. Further complicating matters, Rho and Sanders (2021) found that for some high-skilled groups, it is the higher-earning immigrants who are more likely to leave, which would bias cross-sectional estimates downward. This complexity is echoed in the Korean context by Kim and Lee (2023), who suggest that the apparent "negative assimilation" of non-Asian immigrants (whose wages decline relative to natives over time) may be due to the selective out-migration of the most successful individuals within that group. These findings collectively demonstrate that the observed relationship between credentials and earnings is inextricably linked to the non-random process of return migration. The Primacy of "Immigrant" Status: When Diploma Origin Fails to Matter The central tenet of the user's research—that a domestic degree may have no discernible effect—finds its strongest theoretical support in situations where an individual's macro-identity as "immigrant" or "foreigner" overrides the micro-level signal of their qualifications. Even a gold-plated domestic degree may fail to unlock opportunities if employers' perceptions are dominated by biases associated with a worker's national or ethnic origin. This can occur through several mechanisms. Overt discrimination, rooted in xenophobia or racism, is one possibility. More subtle, but equally potent, is statistical discrimination. This occurs when employers, facing imperfect information about an individual's true productivity, use perceived group averages to make hiring and wage decisions. If an employer holds a preconceived notion—correct or not—that workers from a particular country are, on average, less productive, they may offer a lower wage to an individual from that country, irrespective of their specific qualifications. In this scenario, the signal sent by a domestic degree is simply not strong enough to overcome the powerful, negative signal of group identity. Empirical evidence for this phenomenon is compelling. A study by Støren and Wiers-Jenssen (2010) in Norway provides a particularly stark illustration. By explicitly comparing four distinct groups—natives with domestic degrees, natives with foreign degrees, immigrants with domestic degrees, and immigrants with foreign degrees—they were able to isolate the effects of diploma origin versus ethnic origin. Their key finding was that non-Western immigrants faced a significantly higher risk of unemployment and overqualification regardless of whether their degree was earned in Norway or abroad. This directly suggests that, at least in the Norwegian context, "ethnic origin" was a more salient determinant of labor market outcomes than the "origin of the diploma". This provides a powerful precedent for the hypothesis that in some societies, the benefits of a domestic degree can be nullified by the disadvantages of an immigrant background. Similarly, Arbeit and Warren's (2013) finding that significant wage penalties for foreign degrees persist even after extensive controls suggests that unobserved factors, likely related to discrimination or cultural penalties, are at play. The Interplay of Gender, Source Country, and Host Country Context The effect of a domestic degree is not a universal constant; its value is highly contingent on the individual characteristics of the immigrant and the institutional context of the host country. Lumping all immigrants together into a single analytical category risks masking crucial sources of heterogeneity. Gender is a primary axis of differentiation. Labor markets are not gender-neutral, and immigrant women often face a "double disadvantage." The study by Arbeit and Warren (2013) found not only that the wage penalty for a foreign degree was substantially larger for women than for men (17% vs. 11%), but also that women with foreign degrees faced a higher barrier to employment itself. The decision to participate in the labor force is a more significant source of selection bias for women. It is therefore plausible that the signaling power and economic returns of a domestic degree also differ by gender, intersecting with employers' gendered assumptions about productivity, career commitment, and family responsibilities. The country of origin also fundamentally shapes the value of a domestic credential. The "value-added" of a host-country degree is not uniform across all nationalities. For an immigrant from a country with a highly regarded and structurally similar educational system (e.g., a German immigrant in the U.S.), a domestic degree may offer only a marginal advantage over their foreign credential. Conversely, for an immigrant from a developing nation whose educational system is unfamiliar to host-country employers, a domestic degree can serve as a far more critical bridge across the information gap, acting as an essential tool for validation. Finally, and most critically for the present research, the host country context is paramount. The vast majority of the foundational literature on immigrant assimilation has been conducted in a small number of traditional Western settlement countries, primarily the United States, Canada, and Australia. The institutional frameworks, immigration histories, and cultural norms of these nations are not universal. Extrapolating findings from these contexts to a country like South Korea requires extreme caution. South Korea has a much shorter history as a major immigrant-receiving nation, a far greater degree of ethnic and linguistic homogeneity, and a distinct set of cultural norms governing its labor market. In such a context, it is highly plausible that the social and economic distance between "native" and "foreigner" is more pronounced, and the mechanisms of signaling and trust operate differently. The premium on ethnic insider status may be so high that it renders the distinction between a domestic and foreign credential for an ethnic outsider largely irrelevant.

The relationship between holding a domestic degree and achieving labor market success is far more tenuous and conditional than a superficial reading of the literature might suggest. While the default hypothesis, supported by a significant body of evidence, posits a clear economic premium for host-country qualifications, this effect is not deterministic. This review has demonstrated that the expected advantage can be systematically undermined by a potent combination of factors. First, the data upon which such conclusions are based are often distorted by selection biases, both in terms of who chooses to pursue a foreign education and, more importantly, who chooses to remain in the host country after graduation. The "survivorship bias" inherent in cross-sectional data may lead to a significant overestimation of the true returns to a domestic degree. Second, the theoretical power of a domestic degree as a positive signal can be overwhelmed by the stronger, negative signal of "immigrant" status, particularly for racial or ethnic minorities. As evidence from Norway suggests, when employers prioritize ethnic origin over credential origin, the potential benefits of domestic education can evaporate. Finally, the effect is not uniform but is highly heterogeneous, varying significantly with the immigrant's gender, country of origin, and, critically, the institutional and cultural context of the host country. Therefore, the tentative finding that domestic qualifications offer no clear advantage for foreign workers in the South Korean labor market should not be viewed as an anomaly. Rather, it is an empirically plausible outcome that finds support in established theoretical critiques and specific empirical precedents within the broader literature. It suggests a scenario where the institutional context of Korea—perhaps characterized by strong in-group preferences, high information costs for evaluating any foreign worker, or a labor market segmented along lines of nationality—neutralizes the signaling value of a domestic credential for non-natives. This research, therefore, is well-positioned to make a significant contribution. It serves as a critical test of competing hypotheses in a novel and important context. Does the standard human capital and signaling model, which predicts a premium for domestic degrees, hold in a non-Western, ethnically homogeneous advanced economy? Or does a model emphasizing the primacy of social identity and context-specific discrimination provide a better explanation? By investigating this question, this study can illuminate the specific barriers to integration in the Korean labor market and, more broadly, enrich our understanding of the complex and conditional nature of immigrant economic assimilation worldwide. The ultimate question is not simply if domestic degrees matter, but for whom, where, and under what conditions they are allowed to.

3 Data

The empirical analysis in this study is based on microdata from the Survey on Immigrants' Living Conditions and Labor Force (SILC-LF), a nationally recognized statistical investigation conducted annually by Statistics Korea. This survey represents the most comprehensive and authoritative data source on the foreign population in South Korea, designed specifically to provide foundational evidence for the development of national policies concerning immigrant labor markets, social integration, and multiculturalism. Its relevance has grown in parallel with Korea's rapid transformation into a significant immigrant-destination country. The primary objective of the survey is to produce a detailed, multi-dimensional portrait of immigrants' lives, covering their demographic backgrounds, economic activities, and social adaptation. Operating under the authority of the national Statistics Act, the SILC-LF holds the status of designated statistics, ensuring its methodological rigor and the quality of its data collection processes. The credibility of this dataset makes it an ideal foundation for the sophisticated econometric analysis required to address the nuanced research questions of this paper.

The target population of the survey includes non-Korean nationals and naturalized citizens aged 15 and older who have resided in Korea for a period of at least 91 consecutive days at the time of the survey. This definition encompasses a wide range of immigrants, from temporary foreign workers and international students to long-term residents and new citizens. The survey methodology employs a stratified random sampling approach, with stratification based on all 17 metropolitan areas and provinces across the nation. This ensures a geographically representative sample covering both major urban centers, where immigrant populations are concentrated, and more rural locations. Data collection is primarily conducted through face-to-face interviews administered by trained fieldworkers from regional statistical offices, a method that generally yields higher response rates and data quality compared to other survey modes. For respondents who prefer it, a self-administered questionnaire option is also available. The cross-sectional nature of the survey means that a new random sample of individuals is drawn each year, providing a series of annual snapshots of the immigrant population.

Table 1: Descriptive Statistics of the Sample by Domestic Degree Status

	Full Sample		($d = 1$)		($d = 0$)	
Variable	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Log-Wage, $\ln(y)$	5.5687	0.3399	5.5519	0.4272	5.5696	0.3341
Domestic Degree ($d = 1$)	0.0893	0.2852	1	–	0	–
Employed ($s = 1$)	0.5903	0.4918	0.3621	0.4806	0.6127	0.4871
Job Work Experience						
Under 6 months	0.0949	0.2930	0.1173	0.3218	0.0934	0.2909
6m to 1yr	0.0936	0.2913	0.0827	0.2755	0.0944	0.2923
1yr to 2yrs	0.1646	0.3708	0.1419	0.3490	0.1661	0.3722
2yrs to 3yrs	0.1376	0.3445	0.1145	0.3185	0.1391	0.3461
3+ yrs	0.5093	0.4999	0.5436	0.4981	0.5070	0.5000
Selected Demographic Characteristics						
Female	0.5222	0.4995	0.5345	0.4988	0.5209	0.4996
Married	0.6588	0.4741	0.4513	0.4976	0.6791	0.4668

Note: The table presents the mean and standard deviation of key variables for the full sample and subsamples based on domestic degree status ($d = 1$ for domestic degree holders, $d = 0$ for foreign degree holders). The total number of observations is $N = 118,688$, with $N_{d=1} = 10,603$ and $N_{d=0} = 108,085$.

The temporal scope of our study utilizes the pooled cross-sections of the SILC-LF from 2017 to 2022. This six-year period provides a substantial and robust sample for analysis. Over these years, the survey's scale expanded significantly, reflecting the Korean government's growing recognition of the importance of immigration for its economic and demographic future. In 2017, the survey included 12,650 individuals. This number steadily increased, reaching 23,962 respondents by the 2022 survey wave. By pooling these six annual surveys, we construct a total sample of 118,688 individual observations. This large sample size is a major strength of our study, as it provides sufficient statistical power to conduct detailed analyses, particularly for estimating the effects on smaller subgroups within the immigrant population and for implementing computationally intensive econometric models. The large N also helps in achieving more precise estimates, which is critical for identifying the subtle effects at the heart of our research question. The year 2017 serves as a particularly important starting point for our analysis. It was the first year the survey introduced a crucial question asking whether an immigrant's parents are Korean. This variable is indispensable for our identification strategy, allowing us to construct a key instrumental variable to address the endogeneity of educational choices. Its inclusion enables a more nuanced examination of how family background and early-life context influence subsequent educational and labor market trajectories.

The richness of the SILC-LF lies in its comprehensive questionnaire, which gathers data across multiple domains. For the purpose of this study, we draw upon several key sections. The survey collects detailed demographic information, including age, gender, country of origin (nationality), marital status, number of children, and religion. It also contains a detailed education module, which records the highest level of education completed and, critically for our research, the country in which that degree was obtained. This allows us to construct our key explanatory variable: a binary indicator for holding a domestic (Korean) degree. Furthermore, detailed information on economic and employment status is collected, including job type (e.g., regular vs. temporary position), hours worked, industry, firm size, and coverage by employment and accident compensation insurance.

A defining characteristic of this dataset, which fundamentally shapes our econometric approach, is the way in which wage information is recorded. Instead of collecting income as a precise, continuous variable, the survey measures it in ordered categorical brackets or quantiles. This methodological choice, common in large-scale social surveys to improve response rates for sensitive questions, means that the dependent variable is inherently ordinal. While this precludes the use of standard linear regression models, it makes an Ordered Probit model the most structurally appropriate and statistically sound choice for our analysis. The dependent variable in our model is thus a categorical indicator representing the wage bracket of an employed individual.

To provide a descriptive overview of the pooled 2017-2022 sample, a summary of key characteristics is essential. The sample of 118,688 individuals reflects the diverse nature of immigration to Korea. In terms of national origin, the largest single group consists of ethnic Koreans from China (Joseon-jok), followed by significant populations from Vietnam, non-Korean Chinese, the Philippines, and other Southeast Asian nations. Immigrants from other developed regions such as North America and Europe constitute a smaller but significant portion of the sample. Demographically, the sample is fairly balanced by gender, with a slight majority of male respondents, and the age distribution is concentrated in the prime working years of 30 to 59. A critical variable for our analysis is the source of higher education. Within our sample, we observe that a distinct minority of immigrants with a tertiary education hold a degree from a Korean institution. This variation is key to identifying the premium, or lack thereof, associated with a domestic credential. The employment rate for the sample is substantial, but a significant fraction is either unemployed or out of the labor force, highlighting the importance of correcting for sample selection in our wage models. For those who are employed and report wages, the distribution across the categorical wage brackets is concentrated in the lower-to-middle ranges, underscoring the relevance of studying the factors that facilitate upward economic mobility.

While the SILC-LF is an invaluable resource, a balanced academic assessment requires acknowledging its limitations. The most significant of these is its cross-sectional design. Because the survey interviews a different random sample of individuals each year, it is not possible to track the same person over time. This prevents the use of panel data methods that can control for unobserved time-invariant individual heterogeneity and makes it more challenging to draw strong causal inferences. Our reliance on an instrumental variable strategy is, in part, a direct response to this limitation. A second key limitation is the exclusion of undocumented migrants, particularly those who have overstayed short-term visas. This is a population that likely faces the most precarious labor market conditions, and their absence means that our findings may not generalize to the most vulnerable segments of the immigrant population. Finally, as with any survey based on self-reporting, there is a potential for social desirability bias or underreporting on sensitive topics, such as experiences of discrimination or precise income levels, although the categorical nature of the wage question may mitigate the latter issue to some extent.

Despite these constraints, the strengths of the SILC-LF are substantial and make it uniquely suited for this research. Its large scale, national scope, and the sheer richness of its variables provide an unparalleled opportunity to study the immigrant experience in Korea with a high degree of statistical rigor. The inclusion of both objective indicators (like employment status and firm size) and subjective ones (like life satisfaction and perceived discrimination, though not central to this paper) allows for a holistic understanding of integration. The careful distinction between foreign nationals and naturalized citizens, and the detailed information on visa types, provides further avenues for nuanced analysis. In conclusion, the SILC-LF is the most comprehensive and high-quality dataset available for investigating the economic lives of immigrants in South Korea. Its detailed information on educational origin, employment, and wages, combined with its large sample size, provides a robust empirical foundation upon which our sophisticated econometric model is built, allowing us to credibly identify the effect of a domestic degree while accounting for the complex statistical challenges inherent in the data.

4 Identification Strategy and Estimation

The primary objective of this study is to estimate the causal effect of obtaining a domestic (host-country) degree on the potential wages of foreign workers in South Korea. A naive empirical approach, such as a simple Ordinary Least Squares (OLS) regression of wages on an indicator for domestic education, would be fraught with significant econometric challenges, leading to biased and inconsistent estimates. A credible identification strategy must simultaneously address three fundamental issues inherent in the research question and the available data: (1) the ordinal nature of the dependent variable, as wage data is reported in categorical brackets rather than as a continuous measure; (2) sample selection bias, arising from the fact that wages are only observed for the non-random sub-sample of individuals who are employed; and (3) endogeneity of the key regressor, as the decision to acquire a domestic degree is not random and is likely correlated with unobserved determinants of earnings.

To surmount these challenges, we develop a comprehensive econometric framework based on a sample-selection corrected Ordered Probit model that incorporates an instrumental variable (IV) strategy to account for endogeneity. Given the complexity of the resulting likelihood function, which involves intractable integrals, we employ a Simulated Maximum Likelihood (SML) approach for parameter estimation. Finally, to ensure robust statistical inference that accounts for all sources of uncertainty in this multi-stage estimation process, standard errors are computed using a non-parametric bootstrap procedure. The following subsections detail the theoretical underpinnings of this framework and the practical steps of its implementation.

4.1 Modeling Framework: A Sample-Selection Corrected Ordered Probit

Our modeling approach begins by conceptualizing an unobserved, latent variable, y_i^*, which represents the true potential log-wage of immigrant i. This latent wage is assumed to be a linear function of a vector of individual characteristics, x_i, and an unobserved error term, u_i:

$$y_i^* = x_i'\beta + u_i \tag{1}$$

The vector x_i includes our key variable of interest—an indicator for holding a domestic degree—as well as a comprehensive set of control variables reflecting human capital (e.g., work experience, foreign education level), demographics (e.g., age, gender, marital status), and job characteristics (e.g., firm size, occupation). The parameter vector β represents the returns to these characteristics.

While y_i^* itself is unobserved, we observe the categorical wage bracket, y_i, into which an individual's wage falls. The relationship between the observed category and the latent wage is defined by a set of known, fixed thresholds. A significant advantage of our data is that these thresholds are not parameters to be estimated but are exogenously given by the survey design, corresponding to specific wage values (in our case, the natural logarithm of 100, 200, and 300 thousand KRW). The observed wage category $y_i \in \{1, 2, 3, 4\}$ is thus determined as follows:

$$\begin{aligned} y_i = & \mathbb{1}[y_i^* \leq \ln(100)] + 2 \cdot \mathbb{1}[\ln(100) < y_i^* \leq \ln(200)] \\ & + 3 \cdot \mathbb{1}[\ln(200) < y_i^* \leq \ln(300)] + 4 \cdot \mathbb{1}[\ln(300) < y_i^*] \end{aligned} \tag{2}$$

This structure naturally lends itself to an Ordered Probit model, which correctly handles the ordinal nature of the dependent variable without imposing arbitrary assumptions about the cardinal distances between categories.

The second critical challenge is that wage information is only available for individuals who are employed. The decision to participate in the labor force is a conscious choice, not a random event, and is likely correlated with factors that also determine potential wages. For instance, individuals with unobserved characteristics that lead to higher wages (e.g., greater ambition, stronger work ethic) may also be more likely to seek and find employment. Conversely, individuals who anticipate facing discrimination and low wage offers may be discouraged from participating in the labor market altogether. Ignoring this non-random selection, as a standard Ordered Probit model applied only to the sample of workers would do, leads to sample selection bias, a problem famously articulated by Heckman (1979).

To correct for this bias, we explicitly model the labor force participation decision. We specify a selection equation where an individual's decision to work, denoted by the binary indicator $s_i = 1$, is determined by a separate latent variable, s_i^*, crossing a threshold of zero. This latent variable is a function of a vector of observed characteristics, w_i, and an unobserved error term, v_i:

$$s_i = \mathbb{1}[s_i^* > 0] \quad \text{where} \quad s_i^* = w_i'\gamma + v_i \tag{3}$$

The vector w_i contains variables that influence the labor supply decision, such as marital status, the presence of young children, and other non-labor income sources. For the model to be identified, w_i must contain at least one variable that is not included in the wage equation vector x_i.

The key to correcting for selection bias is the link between the unobserved components of the wage and selection equations. We assume that the error terms, u_i and v_i, follow a bivariate normal distribution with a zero mean vector and a covariance matrix that allows for correlation between them:

$$\begin{bmatrix} u_i \\ v_i \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{bmatrix} \right) \tag{4}$$

Here, the variance of the selection equation error v_i is normalized to 1 for identification, which is standard for probit models. The parameter ρ represents the correlation between the unobserved factors affecting potential wages and those affecting the probability of employment. If ρ is statistically different from zero, it confirms the presence of sample selection bias, and a joint estimation of the two equations is necessary to obtain consistent estimates of β. This structure forms the basis of the sample-selection corrected Ordered Probit model.

4.2 Addressing Endogeneity via Instrumental Variables

The third, and perhaps most challenging, issue is the endogeneity of our key explanatory variable: the indicator for holding a domestic degree. For the coefficient on this variable to be interpreted as a causal effect, the decision to obtain a domestic degree must be uncorrelated with the unobserved determinants of wages, u_i. This assumption is highly unlikely to hold in reality. Unobserved individual traits such as innate ability, ambition, motivation, risk aversion, or access to family resources and social networks are likely to influence both an individual's decision to pursue higher education in Korea and their subsequent earnings potential. For instance, a highly motivated and capable individual may be more likely to both undertake the challenge of a foreign degree program and succeed in the labor market, creating a spurious positive correlation between the domestic degree and wages that is not causal.

To address this endogeneity, we employ an Instrumental Variable (IV) strategy.

A valid instrument is a variable that satisfies two core conditions: (1) relevance, meaning it is strongly correlated with the endogenous variable (the decision to obtain a domestic degree), and (2) the exclusion restriction, meaning it affects the outcome variable (potential wage) only through its effect on the endogenous variable and is not directly correlated with the unobserved error term u_i. Identifying such instruments is a notoriously difficult task. In this study, we propose two instruments based on an immigrant's background prior to their university-level education decision.

Our first instrument is the immigrant's parents' country of origin. The relevance condition is met if the parents' background influences the likelihood of their child studying in Korea. For example, immigrants whose parents are ethnic Koreans, or are from countries with strong historical, cultural, or economic ties to Korea, may face lower information barriers, have access to stronger family support networks, or possess a greater cultural affinity that encourages them to pursue a Korean degree. The exclusion restriction requires that, conditional on the individual's own characteristics (including their own nationality, language proficiency, and time in Korea, which are all controlled for in x_i), their parents' origin does not have a direct causal effect on their present-day wages. The argument is that the influence of parental background on current earnings is primarily channeled through the life-path decisions it shapes, most notably the location of higher education, rather than having a separate, direct impact.

Our second, and arguably more powerful, instrument is an indicator for whether the immigrant was born in South Korea. The relevance of this instrument is exceptionally strong. An individual born in Korea, even if classified as a foreigner for administrative or citizenship purposes (e.g., a 'third-culture kid' or the child of long-term expatriates), is far more likely to be a native speaker of Korean, to be fully integrated into the Korean educational pipeline from a young age, and to possess deep-seated local social networks. These factors create a strong path dependency, dramatically increasing the probability of that individual choosing and successfully matriculating to a Korean university compared to an individual who immigrates later in life. The correlation between being born in Korea and obtaining a domestic university degree is therefore expected to be very high.

The exclusion restriction requires that, conditional on the other covariates in the model, being born in Korea affects an individual's potential wage only through its influence on their educational path. A potential and valid objection is that being born in Korea could have a direct effect on wages by signaling perfect language fluency or a deeper level of cultural assimilation, factors that employers might reward independently of one's university degree. However, our wage equation is carefully specified to control for these potential direct channels. The vector of controls, x_i , includes direct measures of language ability, the total number of years of residence in Korea (which explicitly proxies for general acculturation and the accumulation of country-specific human capital over time), and detailed indicators for current nationality and visa status. The identifying assumption is therefore that, once these mediating factors are accounted for, the primary remaining causal impact of being born in Korea is its powerful influence on the decision to pursue a domestic university degree. Any residual direct effect on wages is assumed to be negligible relative to this strong channeling effect, allowing the birthplace indicator to serve as a valid instrument for identifying the causal impact of the domestic credential itself. By integrating these instruments into our selection-corrected ordered probit framework, we can purge our estimates of the bias introduced by the non-random nature of the educational choice.

4.3 Estimation by Simulated Maximum Likelihood (SML)

The full econometric model, incorporating the ordered probit structure, the sample selection correction, and the instrumental variable approach, results in a complex likelihood function. The probability of observing a specific outcome for individual i (e.g., $y_i = j$ and $s_i = 1$) is derived by integrating the conditional probabilities over the distribution of the unobserved heterogeneity. This leads to an expression for the likelihood contribution of each individual that contains a one-dimensional integral, as shown in the model specification. For example, the joint probability for an employed individual in the first wage category is:

$$\Pr(y_i = 1, s_i = 1 \mid x_i, w_i) = \int_{-w_i'\gamma}^{\infty} \Phi\left(\frac{\ln(100) - x_i'\beta - \theta v}{\sigma_e}\right) \phi(v) dv \tag{5}$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ are the standard normal CDF and PDF, respectively, and the parameters θ and σ_e are functions of the underlying structural parameters ρ and σ.

A critical obstacle to estimation is that this integral, and the corresponding integrals for the other wage categories, do not have a closed-form analytical solution. This intractability renders standard Maximum Likelihood Estimation (MLE), which would require direct evaluation of the likelihood function, infeasible. To overcome this, we turn to a powerful computational technique: Simulated Maximum Likelihood (SML). The intuition behind SML is straightforward: if an integral representing an expected value cannot be solved analytically, it can be approximated with arbitrary accuracy by averaging the function inside the integral over a large number of pseudo-random draws from the relevant distribution.

The SML estimation procedure for our model unfolds as follows. First, as our model contains a selection equation, we can obtain a consistent estimate of the selection parameters, $\hat{\gamma}$, from a first-stage Probit regression of the employment indicator s_i on the variables in w_i. This is a standard approach in two-step selection models and provides the necessary values for the lower bound of integration, $-w_i'\hat{\gamma}$.

Next, we construct the simulated log-likelihood function. For each observation i in the sample, we approximate the intractable integral that defines its probability contribution. This is done by taking a large number, H, of independent draws, z_h (for $h = 1, \ldots, H$), from a standard univariate normal distribution. The expectation is then approximated by a sample mean. For instance, the probability for an individual in wage category j is simulated as:

$$
\begin{aligned}
&\Pr(y_i = j, s_i = 1) \\
\approx &\frac{1}{H}\sum_{h=1}^{H}\left[\left\{\Phi\left(\frac{a_j - x_i'\beta - \theta z_h}{\sigma_e}\right) - \Phi\left(\frac{a_{j-1} - x_i'\beta - \theta z_h}{\sigma_e}\right)\right\}\cdot \mathbb{1}[z_h > -w_i'\hat{\gamma}]\right]
\end{aligned}
\tag{6}
$$

where a_j represents the relevant wage threshold (ln(100), etc.), and the indicator function $\mathbb{1}[\cdot]$ ensures that the average is taken only over draws that satisfy the selection condition (i.e., correspond to being employed). The overall simulated log-likelihood, $\ln \mathcal{L}_{SML}$, is the sum of the logarithms of these simulated probabilities across all individuals in the sample.

Finally, we use a numerical optimization algorithm (specifically, L-BFGS-B) to find the parameter vector $\xi = (\beta, \theta, \sigma_e)$ that maximizes this simulated log-likelihood function. Once the optimal estimates $(\hat{\beta}, \hat{\theta}, \hat{\sigma}_e)$ are obtained, the underlying structural parameters of the covariance matrix can be recovered as $\hat{\sigma} = \sqrt{\hat{\theta}^2 + \hat{\sigma}_e^2}$ and $\hat{\rho} = \hat{\theta}/\hat{\sigma}$.

4.4 Inference via Non-Parametric Bootstrap

Obtaining accurate standard errors for the estimated parameters in such a complex, multi-stage, simulation-based model is a non-trivial task. While analytical formulas for the variance-covariance matrix (such as those based on the Hessian or the outer product of the gradient) can be derived in simpler models, their application here is fraught with complications. An analytical approach would need to account for multiple sources of uncertainty simultaneously: the sampling variation from the first-stage estimation of $\hat{\gamma}$, the simulation error introduced by approximating the likelihood function, and the standard sampling variability from the final optimization step. This makes such formulas exceedingly complex and reliant on potentially strong assumptions.

Given these challenges, we adopt a **non-parametric bootstrap** procedure to compute the standard errors. This method is computationally intensive but provides a robust and reliable way to estimate the sampling distribution of our parameters, as it inherently accounts for all sources of variation in the estimation process without imposing restrictive assumptions. The procedure is implemented as follows:

1. We draw a "bootstrap sample" of size N by resampling with replacement from our original dataset of N observations.

2. On this bootstrap sample, we re-run the *entire* estimation procedure from start to finish: we first estimate the first-stage probit model to obtain a new $\hat{\gamma}_b$, and then we use SML to estimate the main model parameters, yielding a bootstrap estimate vector $\hat{\xi}_b = (\hat{\beta}_b, \hat{\theta}_b, \hat{\sigma}_{e,b})$.

3. We repeat this process a large number of times, B (e.g., $B = 500$), generating a distribution of B different parameter vectors.

4. The standard error for any given parameter (e.g., the coefficient on domestic degree) is then calculated as the standard deviation of its empirical distribution across these B bootstrap replications.

This bootstrap approach provides a credible and robust foundation for statistical inference, allowing us to accurately assess the statistical significance of our findings in the context of our sophisticated and computationally demanding econometric model.

5 Result

This section presents the empirical findings from our investigation into the wage premium of a domestic degree for foreign workers in South Korea. The analysis unfolds in two stages. First, we examine the coefficient estimates from our primary econometric model—the sample-selection corrected Ordered Probit with instrumental variables, estimated via Simulated Maximum Likelihood (SML). This allows us to understand the underlying structural relationships between our variables of interest and the latent potential wage, controlling for the econometric challenges of selection and endogeneity. Second, we translate these coefficients into Average Partial Effects (APEs). This subsequent analysis provides a more intuitive interpretation of the results by quantifying how a change in a given variable affects the predicted probability of an immigrant falling into each specific wage category. Together, these two sets of results move beyond a simplistic assessment and reveal a nuanced, dynamic relationship between domestic education, work experience, and immigrant wage assimilation. While a cursory look at the main effect of a domestic degree might suggest a negligible impact, a deeper, more careful interpretation of the interaction terms and partial effects unveils a more complex and compelling narrative: a domestic degree acts as a latent asset whose economic value is conditional upon, and activated by, the accumulation of host-country work experience.

5.1 Coefficient Estimates from the Simulated Maximum Likelihood Model

The primary estimation results from our preferred specification, the SML model, are presented in Table 1. This table provides the estimated coefficients that constitute the core of our structural model, revealing the directional and statistical significance of various factors on the latent log-wage variable. For comparative purposes, we also present the results from a standard Heckman two-step model, though our main discussion will focus on the more robust SML estimates which simultaneously address the ordinal nature of the dependent variable and the selection issue.

The analysis begins with our central variable of interest: the indicator for holding a domestic degree, denoted as domestic degree (d). The estimated coefficient on this variable in the SML model is 0.0091, with a corresponding t-statistic of 0.76. This result is the linchpin of our initial, though incomplete, conclusion. The coefficient is small in magnitude and, more importantly, is statistically indistinguishable from

Table 2: Comparison of Heckman and Simulated ML Estimation Results

	Heckman			Simulated ML		
Variable	Coefficient	Std. Err.	t-value	Coefficient	Bootstrap S.E.	t-value
domestic degree (d)	−0.0153	0.0129	−1.19	0.0091	0.0120	0.76
Job Work Experience						
6m to 1yr (exper2)	0.0378	0.0047	8.04	0.0198	0.0046	4.32
1yr to 2yrs (exper3)	0.0497	0.0042	11.76	0.0326	0.0040	8.13
2yrs to 3yrs (exper4)	0.0696	0.0044	15.87	0.0510	0.0041	12.38
3+ yrs (exper5)	0.0986	0.0039	25.51	0.0844	0.0038	22.42
d*exper2	0.0334	0.0193	1.73	0.0149	0.0190	0.78
d*exper3	0.0306	0.0168	1.82	0.0079	0.0172	0.46
d*exper4	0.0056	0.0181	0.31	0.0004	0.0187	0.02
d*exper5	0.0494	0.0143	3.45	0.0366	0.0144	2.54
Education Level						
Middle School	0.0174	0.0041	4.22	0.0176	0.0036	4.88
High School	0.0321	0.0038	8.37	0.0340	0.0034	10.06
2yr College +	0.0489	0.0043	11.38	0.0640	0.0040	15.87
Demographics and Job Characteristics						
30s	0.0294	0.0030	9.91	0.0336	0.0028	12.17
40s	0.0369	0.0039	9.49	0.0500	0.0035	14.20
50s	0.0096	0.0044	2.20	0.0411	0.0040	10.24
60s	−0.0492	0.0052	−9.39	−0.0211	0.0050	−4.25
Female=1	−0.1608	0.0024	−67.82	−0.1676	0.0025	−67.04
Married=1	0.0069	0.0027	2.53	−0.0066	0.0027	−2.42
hasChildren=1	0.0059	0.0029	2.04	−0.0098	0.0030	−3.22
Regular Position=1	0.0286	0.0027	10.41	0.0392	0.0028	13.95
Employment Insurance=1	−0.0022	0.0026	−0.85	0.0038	0.0029	1.30
Accident Comp. Insurance=1	0.0500	0.0030	16.51	0.0537	0.0033	16.37
Firm Size						
5–9	0.0414	0.0033	12.38	0.0334	0.0037	8.95
10–29	0.0732	0.0032	23.02	0.0630	0.0035	18.03
30–49	0.0934	0.0040	23.29	0.0860	0.0045	19.30
50–299	0.1108	0.0035	31.67	0.1080	0.0040	26.94
300+	0.1833	0.0061	29.96	0.1676	0.0075	22.30
Model Specifics						
constant	4.9587	0.0095	524.09	5.2422	0.0094	555.43
imr	0.1899	0.0150	12.65	-	-	-
θ	-	-	-	0.0182	0.0040	4.51
σ_e	-	-	-	0.1843	0.0013	139.20
Survey Year FE		Yes			Yes	
Nationality		Yes			Yes	
Visa		Yes			Yes	

Note: This table compares estimation results from the Heckman two-step model and the Simulated Maximum Likelihood (SML) model. The dependent variable is the log wage.

[a] Nationality includes 15 categories: naturalized Koreans, Korean-Chinese, non-Korean Chinese, Vietnamese, Uzbeks, Filipinos, Indonesians, Japanese, Thais, Mongolians, other Asian, U.S./Canada, Europe, Oceania, and others.

[b] Visa includes 8 categories: E-9, H-2, E-1/E-3, F-4, F-5, F-2-1/F-6, D-2/D-4-1/D-4-7, and others.

zero at any conventional level of significance. A naive interpretation of this single coefficient would suggest that, after controlling for a host of other characteristics, obtaining a degree from a Korean university provides no discernible wage premium for a foreign worker. This finding, on its own, appears to support the hypothesis that the value of a domestic credential may be muted or nullified within the specific context of the Korean labor market, where factors such as nativity or social identity might override formal qualifications.

However, such a conclusion would be premature and ultimately incorrect due to the inclusion of interaction terms in our model specification. In a model where a variable is interacted with others, its main effect coefficient does not represent the average effect across the entire sample. Instead, it represents the effect only for the baseline reference group—that is, the group for whom all interacted variables are equal to zero. In our model, the domestic degree variable is interacted with categories of work experience. The omitted reference category for experience is "Under 6 months." Therefore, the statistically insignificant coefficient of 0.0091 should be interpreted precisely as the effect of a domestic degree for a foreign worker with virtually no (less than six months) work experience in Korea. This finding is, in itself, economically plausible. For a recent graduate or a newcomer to the labor market, the tangible benefits of a degree—whether it be specialized knowledge or professional networks—may require time to materialize into higher productivity and, consequently, higher wages. Employers may be hesitant to offer a wage premium based on a credential alone without demonstrated evidence of on-the-job competence.

The true narrative of the value of a domestic degree is revealed through the coefficients of the interaction terms. While the interactions with shorter-term experience (d*exper2, d*exper3, d*exper4) are themselves not statistically significant, the coefficient for the interaction with the highest level of experience, d*exper5 (representing 3+ years of experience), is 0.0366 with a robust t-statistic of 2.54. This result is highly significant and positive, and it is the key to unlocking the full story. It provides strong evidence that the effect of a domestic degree is not constant across all immigrants; rather, it is conditional on their level of work experience. The positive coefficient indicates that the wage premium for a domestic degree becomes significantly larger as an immigrant accumulates substantial experience in the Korean labor market.

The total effect of a domestic degree for an experienced worker is therefore the sum of the main effect and the relevant interaction effect. For an immigrant with three or more years of experience, the total effect on their latent log-wage is the sum of the coefficient for d and the coefficient for d*exper5: 0.0091+0.0366=0.0457. This combined effect is not only positive and meaningful in magnitude, suggesting an approximate 4.6 percentage point increase in potential wages, but its statistical significance is strongly implied by the significance of the interaction term itself. This demonstrates that while a domestic degree may be a dormant asset upon labor market entry, it becomes an active and valuable one over time. It is not that the degree is worthless; its value is unlocked through the channel of sustained workplace participation.

Turning to the other covariates in the model, the results are largely consistent with standard predictions from labor economics and human capital theory. The direct effects of work experience (exper2 through exper5) show a clear, monotonically increasing relationship with wages, confirming that experience is highly valued. Similarly, higher levels of general education (High School, 2yr College +) are associated with statistically significant wage premia, as expected. The demographic variables reveal important patterns of stratification within the immigrant labor market. The coefficient for Female is large, negative, and highly significant (-0.1676), indicating a substantial gender wage gap among foreign workers, even after controlling for a wide range of productivity-related characteristics. The coefficients for being Married and having hasChildren are negative and statistically significant in the SML model, suggesting potential wage penalties associated with family status, particularly for women who often bear a disproportionate share of domestic responsibilities. The age profile exhibits the classic inverted U-shape, with wages peaking for workers in their 40s and declining thereafter.

Job characteristics also play a significant role. Holding a Regular Position is associated with a significant wage premium compared to temporary or daily work, reflecting greater job stability and better compensation packages. Perhaps most strikingly, Firm Size emerges as a powerful determinant of wages. There is a clear and monotonic wage premium associated with working in larger firms, with employees in firms of 300 or more workers earning a substantial premium over those in very small firms (the reference category). This is consistent with a large body of literature documenting the firm-size wage effect, which is often attributed to factors like rent-sharing, better technology, and more selective hiring in larger enterprises.

Finally, the parameters specific to our selection model provide justification for our methodological approach. The parameter theta, which is a function of the correlation (rho) between the unobserved determinants of employment and wages, is estimated to be 0.0182 with a t-statistic of 4.51. Its statistical significance confirms the presence of a positive selection bias: unobserved factors that make an immigrant more likely to be employed are also positively correlated with their potential wage. This finding validates our decision to employ a sample-selection corrected model, as ignoring this correlation would have led to biased and inconsistent estimates of the returns to education and other characteristics.

5.2 Average Partial Effects on Wage Probabilities

While the coefficients from the SML model are crucial for understanding the structural relationships in our model, their interpretation can be abstract as they relate to an unobserved latent variable. To provide a more intuitive and practical understanding of our findings, we calculate the Average Partial Effects (APEs). The APEs, presented in the subsequent table, translate the estimated coefficients into tangible changes in the predicted probability of an individual falling into each of the four observable wage categories. This analysis allows us to see not just the direction and significance of an effect, but also its real-world magnitude in terms of labor market outcomes.

The APE analysis provides the clearest and most compelling illustration of our central narrative. It confirms that the value of a domestic degree is dynamic and

0.9

Table 3: Average Partial Effects (APE) on Log Wage Probabilities with Bootstrap Standard Errors

	$\Pr(\mathrm{w} \leq \ln(100))$		$\Pr(\ln(100) < \mathrm{w} \leq \ln(200))$		$\Pr(\ln(200) < \mathrm{w} \leq \ln(300))$		$\Pr(\mathrm{w} > \ln(300))$	
Variable	APE	S.E.	APE	S.E.	APE	S.E.	APE	S.E.
Domestic Degree (d)	0.0000	0.0003	-0.0033	0.0001	-0.0196	0.0000	0.0230	0.0004
Interaction with Experience:								
$d \times$ exper2	0.0000	0.0003	-0.0038	0.0003	-0.0140	0.0001	0.0177	0.0002
$d \times$ exper3	0.0000	0.0006	-0.0020	0.0000	-0.0074	0.0004	0.0095	0.0010
$d \times$ exper4	0.0000	0.0005	-0.0001	0.0000	-0.0004	0.0004	0.0004	0.0009
$d \times$ exper5	-0.0001	0.0005	-0.0089	0.0001	-0.0338	0.0004	0.0427	0.0010

deeply intertwined with work experience, primarily functioning as a catalyst for upward economic mobility. Let us first focus on the highest wage bracket, Prob(wage ¿ ln(300)), which represents the probability of earning a high income (approximately 3 million KRW per month or more). The main effect of Domestic Degree (d) shows an APE of 0.0230. This means that, for an immigrant with minimal work experience, holding a domestic degree increases the probability of being in this top wage category by a modest but statistically significant 2.3 percentage points. This effect, on its own, is positive.

However, the truly powerful effect is revealed through the interaction term with experience. The APE for $d \times$ exper5 is 0.0427. This indicates that for an immigrant with three or more years of experience, holding a domestic degree provides a substantial additional boost, increasing their probability of securing a high-paying job by a further 4.3 percentage points. This is a large and economically meaningful effect. It shows that the combination of a domestic credential and proven work experience is a powerful driver of success in the Korean labor market.

The other side of this story is told by looking at the APEs for the lower and middle wage brackets. Here, the partial effects for the domestic degree and its interactions are consistently negative. For example, in the ln(200) ¡ wage ¡= ln(300) category, the APE for d is -0.0196 and for d x exper5 is -0.0338. This does not mean the degree is "useless" or has a negative impact on workers in this category. Rather, it means that holding a domestic degree significantly decreases the probability of an individual being "stuck" in these lower- and middle-tier wage brackets. The credential acts as an elevator, reducing the likelihood of remaining on the lower floors and increasing the likelihood of reaching the top floor. This is the hallmark of a variable that promotes upward mobility. The domestic degree provides a pathway out of lower-paying work and into higher-paying careers, and this pathway becomes much wider and easier to traverse once an individual has established a foothold in the labor market through experience.

In summary, the results from both the SML coefficient estimates and the Average Partial Effects analysis, when interpreted together, paint a consistent and sophisticated picture. The initial hypothesis, formed from a cursory glance at the insignificant main effect of the domestic degree, is shown to be an oversimplification. A domestic degree is not a universally applied premium, nor is it worthless. Instead, our findings suggest it functions as a conditional asset. For immigrants just starting their careers in Korea, its immediate monetary value is limited. However, for those who remain and accumulate local work experience, the domestic degree becomes a key differentiator, significantly enhancing their ability to achieve high earnings and avoid low-wage traps. The conclusion is not that domestic education fails, but that its success is contingent on its integration with practical, on-the-job experience. It is the synergy between the formal credential and demonstrated workplace competence that ultimately unlocks the full economic potential of foreign workers in the Korean labor market.

5.2.1 Computation of Average Partial Effects and Standard Errors

While the coefficients from the SML estimation provide valuable insights into the direction and statistical significance of the model's structural parameters, they are not directly interpretable in a practical sense. The coefficients represent the effect of a variable on the unobserved, latent log-wage index (y_i^*), not on the actual probability of observing an individual in a specific wage category. To translate our findings into more meaningful and intuitive quantities, we compute the Average Partial Effects (APEs). The APE measures the average change in the predicted probability of a particular outcome—in our case, falling into a specific wage bracket—resulting from a one-unit change in an explanatory variable, holding all other variables constant. This approach allows us to answer practical questions, such as: "On average, how does holding a domestic degree change an immigrant's probability of earning more than 3 million KRW per month?"

The calculation of the APEs in the context of our complex model—which is non-linear, involves sample selection, and is estimated via simulation—requires a multi-step process. The fundamental building block is the predicted probability for each individual for each possible outcome. Let $\xi = (\beta, \theta, \sigma_e)$ represent the vector of parameters estimated from our SML model. The predicted joint probability that individual i is employed ($s_i = 1$) and falls into wage category j (where $j \in \{1, 2, 3, 4\}$), given their characteristics x_i and w_i, is given by the integral that we previously established as intractable. We approximate this probability using Monte Carlo integration with H random draws, as implemented in our SML procedure. This simulated probability, which we denote as $\hat{\Pr}(y_i = j, s_i = 1 \mid x_i, w_i; \xi)$, is calculated as:

$$\frac{1}{H}\sum_{h=1}^{H}\left[\left\{\Phi\left(\frac{a_j - x_i'\beta - \theta z_h}{\sigma_e}\right) - \Phi\left(\frac{a_{j-1} - x_i'\beta - \theta z_h}{\sigma_e}\right)\right\} \cdot \mathbb{1}[z_h > -w_i'\hat{\gamma}]\right] \quad (7)$$

where $z_h \sim \mathcal{N}(0, 1)$ are the pseudo-random draws, a_j are the known log-wage thresholds (with $a_0 = -\infty$ and $a_4 = \infty$), $\hat{\gamma}$ is the vector of coefficients from the first-stage selection model, and $\mathbb{1}[\cdot]$ is the indicator function that ensures the averaging occurs only over draws corresponding to the selection-into-employment condition. The conditional probability of being in wage category j given employment is then

$\hat{\Pr}(y_i = j \mid s_i = 1, x_i, w_i; \xi) = \hat{\Pr}(y_i = j, s_i = 1 \mid \dots)/\Phi(w_i'\hat{\gamma})$.

With this ability to compute individual-level predicted probabilities, the APE for a discrete (binary) explanatory variable, say the k-th variable in the vector x_i, denoted d_k, is calculated through the following steps. First, for each individual i in our sample ($i = 1, \dots, N$), we create two counterfactual scenarios. In the first scenario, we set the value of the variable d_k to 0, holding all other characteristics in x_i and w_i at their observed values. We denote this counterfactual vector of characteristics as $x_i^{(k,0)}$. In the second scenario, we set d_k to 1, creating the vector $x_i^{(k,1)}$. It is crucial to note that when the variable of interest is involved in interaction terms, as is the case for our domestic degree variable, all corresponding interaction terms are re-calculated based on the counterfactual value of d_k.

Second, using the estimated parameter vector ξ, we compute the predicted conditional probability of being in wage category j for each individual under both scenarios. Let these be $\hat{\Pr}(y_i = j \mid s_i = 1, x_i^{(k,0)}, w_i; \xi)$ and $\hat{\Pr}(y_i = j \mid s_i = 1, x_i^{(k,1)}, w_i; \xi)$. The individual-specific Partial Effect (PE) for individual i is the difference between these two probabilities:

$$\mathrm{PE}_{ij,k}(\xi) = \hat{\Pr}(y_i = j \mid s_i = 1, x_i^{(k,1)}, w_i; \xi) - \hat{\Pr}(y_i = j \mid s_i = 1, x_i^{(k,0)}, w_i; \xi) \quad (8)$$

This value represents how the probability of individual i being in wage category j changes when their status on variable d_k switches from 0 to 1.

Finally, the Average Partial Effect (APE) is simply the average of these individual-specific effects taken over the entire sample of N individuals who are in the labor force:

$$\mathrm{APE}_{j,k}(\xi) = \frac{1}{N} \sum_{i=1}^{N} \mathrm{PE}_{ij,k}(\xi) \quad (9)$$

This procedure is performed for each variable of interest and for each of the four wage categories, yielding the comprehensive set of results presented in our APE table. The point estimates of the APEs reported are calculated using the main parameter estimates from our SML model, $\hat{\xi}$.

To conduct statistical inference on these APEs, we require standard errors. Given the complexity of the APE calculation—which involves non-linear functions of parameters that were themselves estimated via a multi-stage simulation-based method—deriving analytical standard errors is practically infeasible. Therefore, we rely on a non-parametric bootstrap procedure, which provides a robust and computationally tractable method for estimating the sampling variability of our APEs. This procedure fully accounts for all sources of uncertainty, including the estimation of γ in the selection model and the SML estimation of ξ.

The bootstrap process for obtaining the standard error of an APE is as follows. Let $\{\hat{\xi}_b\}_{b=1}^{B}$ be the set of B parameter vectors obtained from the bootstrap procedure described in the previous section, where each $\hat{\xi}_b$ is the result of applying the full SML estimation to the b-th bootstrap sample. For each of these B parameter vectors, we re-calculate the Average Partial Effect for variable k and outcome category j, yielding a distribution of B APEs:

$$\left\{\text{APE}_{j,k}(\hat{\xi}_1), \text{APE}_{j,k}(\hat{\xi}_2), \ldots, \text{APE}_{j,k}(\hat{\xi}_B)\right\} \tag{10}$$

The bootstrap standard error for our APE point estimate is then simply the sample standard deviation of this empirical distribution of bootstrapped APEs:

$$\widehat{\text{S.E.}}(\text{APE}_{j,k}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\text{APE}_{j,k}(\hat{\xi}_b) - \overline{\text{APE}}_{j,k}\right)^2} \tag{11}$$

where $\overline{\text{APE}}_{j,k}$ is the mean of the B bootstrapped APE values. This robust method of inference ensures that our conclusions about the statistical significance of the partial effects are based on a credible and comprehensive assessment of their sampling uncertainty.

6 Conclusion

This study embarked on an investigation into a question of central importance to the economics of immigration and the formulation of evidence-based policy: does the acquisition of a domestic degree confer a tangible economic premium upon foreign workers in a non-traditional, ethnically homogeneous host country? Set against the backdrop of South Korea's rapid demographic transformation, this research sought to move beyond the well-established findings from Western settlement nations and test the universality of human capital and signaling theories in a new and distinct institutional context. The foundational premise of a vast body of literature suggests that host-country qualifications should act as a powerful vehicle for economic assimilation, serving both to build locally relevant skills and to signal productivity to employers operating under information asymmetry. Our empirical investigation, however, uncovers a far more nuanced and conditional reality. The findings of this paper challenge a simplistic interpretation of the value of credentials, suggesting that a domestic degree is not an automatic passport to higher earnings. Instead, it functions as a latent asset, a form of potential energy whose economic value is unlocked and activated only through the catalyst of sustained, on-the-job experience within the host country's labor market. This conclusion not only refines our understanding of immigrant integration in Korea but also offers broader implications for how we conceptualize the interplay between formal education, practical experience, and social identity in shaping economic outcomes.

Our empirical journey began by confronting the inherent econometric challenges posed by the research question and the available data. Recognizing that a naive comparison of wages would be confounded by the ordinal nature of the income data, the non-random selection of individuals into employment, and the endogeneity of the educational choice itself, we constructed a robust analytical framework. This framework, centered on a sample-selection corrected Ordered Probit model with instrumental variables and estimated via Simulated Maximum Likelihood, was designed to isolate the causal effect of a domestic degree on the potential wages of foreign workers. The results yielded by this sophisticated model were, at first glance, perplexing and seemingly contradictory. The primary coefficient for the domestic degree variable was statistically indistinguishable from zero, appearing to lend credence to the initial hypothesis that the credential's signal might be muted or ignored entirely in the Korean context. This initial result suggested a labor market where the macro-identity of "foreigner" could potentially override the micro-level qualification of holding a domestic degree, a finding that would align with studies like Støren and Wiers-Jenssen (2010) which found ethnic origin to be a more salient predictor of labor market mismatch than the origin of a diploma in Norway.

However, a deeper and more careful interpretation, one that fully considers the specified interaction between education and experience, reveals a different and more compelling narrative. The statistically insignificant main effect of the domestic degree does not represent its average effect, but rather its effect for the specific reference group of immigrants with virtually no work experience in Korea. The true story unfolds through the highly significant and positive coefficient on the interaction term between a domestic degree and substantial (3+ years) work experience. This finding is the cornerstone of our contribution. It demonstrates unequivocally that the economic return to a domestic degree is not a static constant but a dynamic variable, conditional upon an immigrant's tenure in the workplace. The total effect for an experienced foreign worker is positive, statistically significant, and economically meaningful. This resolves the apparent contradiction: the domestic degree is not worthless; its value is simply deferred, requiring a period of practical application before it is fully recognized and rewarded by the labor market.

This structural finding was further illuminated and confirmed by our analysis of Average Partial Effects (APEs). By translating the abstract coefficients into tangible changes in predicted probabilities, the APEs provided an intuitive depiction of the mechanism at play. The results showed that holding a domestic degree, particularly when combined with significant work experience, acts as a powerful catalyst for upward economic mobility. It substantially increases the probability of an immigrant securing a position in the highest wage category while simultaneously decreasing the probability of them being confined to the lower- and middle-income brackets. The credential, therefore, functions as an elevator, but one that requires the key of sustained employment to operate. This synergy between formal education and practical experience suggests that the value of a domestic degree lies not just in the signal it sends at the point of hiring, but in its capacity to enhance the productivity and adaptability of a worker over the long term, making their accumulated experience more valuable than it would have been otherwise.

These findings have profound implications for the canonical theories of labor economics. They compel a refinement of both human capital and signaling theories as they apply to immigrant populations. A simplistic human capital model might predict an immediate return to the skills acquired in a domestic program. Our results suggest that this human capital, while real, may require a "validation period" in the workplace before its productivity-enhancing qualities are fully realized and compensated. Similarly, a basic signaling model would posit that a domestic degree sends a clear, immediate signal of quality to overcome information asymmetry. Our findings indicate that this signal, while present, may be initially weak or heavily discounted by employers, especially in a context of high cultural or social distance. The signal only gains its full strength when it is corroborated by a track record of reliable on-the-job performance. Our results thus point towards a hybrid model, where the credential acts as an initial, albeit imperfect, signal that is subsequently validated and amplified by the accumulation of observable, experience-based human capital. This dynamic interplay offers a more realistic portrayal of the assimilation process than a model that treats education and experience as purely additive components.

The policy implications stemming from this research are significant, particularly for South Korea and other emerging immigration destinations. The primary takeaway is that policies aimed at facilitating immigrant integration cannot be one-dimensional. A strategy focused solely on encouraging or subsidizing access to domestic higher education, while well-intentioned, is likely to be insufficient and may even lead to disillusionment among foreign graduates who fail to see immediate returns on their substantial investment of time and resources. Our findings advocate for a more holistic, two-pronged approach. The first prong is indeed to ensure access to quality domestic education and to facilitate the recognition of foreign credentials. The second, and equally critical, prong is to create robust institutional pathways that connect these graduates to meaningful, long-term employment where they can activate the latent value of their degrees.

This could involve a suite of targeted policies. For instance, government-supported internship or co-op programs specifically for foreign students could provide the crucial first step into the professional labor market. Reforming post-graduation work visa schemes to provide longer and more flexible pathways to permanent employment would encourage both immigrants and employers to invest in long-term relationships. Furthermore, programs designed to bridge the gap between academia and industry—such as mentorship initiatives connecting senior foreign professionals with recent graduates, or workshops on the specific nuances of the Korean corporate culture—could help mitigate the initial frictions that appear to dampen the returns to education. Ultimately, the policy goal should shift from simply producing foreign graduates to fostering experienced foreign professionals.

This study, like all empirical research, is subject to limitations that open avenues for future inquiry. The most significant constraint is the cross-sectional nature of the SILC-LF data. While our instrumental variable and sample selection corrections are designed to mitigate the associated econometric problems and get closer to a causal interpretation, they cannot fully substitute for true longitudinal data. The ability to track the same individuals over time, from their point of graduation through their careers, would allow for a more direct and powerful analysis of wage trajectories, providing a cleaner separation of assimilation effects from unobserved cohort differences, a point powerfully made by scholars like Lubotsky (2007). The future availability of such panel data in Korea would represent a major step forward for research in this field.

Furthermore, our analysis treats the "domestic degree" as a monolithic category. Future research could achieve greater nuance by exploring heterogeneity within this group. Do the returns to a domestic degree vary by field of study? Does an engineering or medical degree from a top Korean university have a more immediate and powerful impact than a humanities degree from a less prestigious institution? How do these effects differ by the immigrant's specific country of origin, interacting with pre-existing stereotypes or cultural affinities? Answering these questions would provide a more fine-grained understanding of the mechanisms at play. Finally, while our quantitative model provides a robust picture of the outcomes, it cannot fully illuminate the processes behind them. Qualitative research, including in-depth interviews with human resource managers, corporate executives, and foreign graduates themselves, would be invaluable for uncovering the on-the-ground realities of hiring decisions, workplace integration, and the lived experience of navigating the Korean labor market as a credentialed foreigner.

In conclusion, this paper contributes to the global discourse on immigration by providing a detailed case study from a crucial, under-researched context. By demonstrating the conditional nature of the returns to a domestic degree, we challenge the notion of a universal, automatic premium for host-country education. The central finding—that a domestic degree in South Korea operates as a latent asset activated by local work experience—offers a more dynamic and realistic model of economic assimilation. It suggests that the path to integration is not a simple transaction of exchanging a credential for a wage, but a process of synergy, where formal knowledge and practical application must work in concert. For foreign workers seeking to build a future in Korea, and for the policymakers tasked with creating a fair and prosperous multicultural society, the message is clear: education is a vital key, but the door to economic success is only fully opened once that key is turned by the hand of experience.

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